



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 03.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : CHE F211
Course Title : Chemical Process Calculations
Instructor-in-Charge : HARE KRISHNA MOHANTA
Tutorial Instructor(s): Srinivas Appari, Rahul Kumar (RS)

1. Course Description:

Historical overview of Chemical Engineering, Principles of balancing with examples to illustrate differential and integral balances lumped and distributed balances, Material balances in simple systems involving physical changes and chemical reactions, Systems involving recycle, purge and bypass, Properties of substances: single component & multicomponent, single and multiphase systems. Ideal liquid and gaseous mixtures, Energy balance calculations in simple systems, Introduction to Computer aided calculations-steady state material and energy balances for chemical plants

2. Scope and Objective of the Course:

This course provides an introduction to the basic fundamentals of chemical engineering. It deals with calculations, which form the basis of the design of a chemical industrial plant or process. It aims to inculcate conceptual knowledge as well as systematic and sound problem solving skills in the student. The concept of material and energy balances are introduced, explained, and worked with the help of examples from the real world chemical process plants. Fundamentals of this course would be required for other courses like Chemical Process Technology, Chemical Reaction Engineering, Separation Processes and Process Modelling and Simulation.

3. Course Learning Outcomes (CLO):

CLO1. At the end of this course, the students must be able to apply important definitions and conventions in chemical processes.

CLO2. They can solve material and energy balances for single and multiple units with or without chemical reaction and feedback/bypass.

3. Text Books (TB): Himmelblau, D.M. & Riggs, J.B. "Basic Principles and Calculations in Chemical Engineering", PHI, 8th Edition 2012.

4. Reference Books (RB): Felder R.M., Rousseau, R.W. "Elementary Principles of Chemical Processes" John Wiley & Sons, 3rd Edition 2000.

5. Method of Conduct of the Course:

The course Chemical Process Calculation is organized into three weekly lectures of 50 minutes each, complemented by tutorials, self-study, and continuous assessment. Weekly tutorials emphasize numerical problem-solving to strengthen analytical skills. Independent study is encouraged through assigned readings, problem sets, and use of our Learning Management System (NALANDA) for lecture notes, resources, and discussion forums.

6. Course Plan:

Week/Module No. and Title	Topics Covered and Lecture Numbers	Reference (TB)	Learning Outcomes
M1: Introductory Concepts	L1-L3: Introduction to Chemical Engineering; Importance of CPC; Units & Dimensions; Conversion of Units; Dimensional consistency; Mole, Density, Specific Gravity, Flow Rates, Concentration	Ch. 2.1–2.12	Recognize role of CPC; review preliminaries; apply basic definitions in calculations



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

M2: Material Balance preliminaries	L4-L6: Open vs Closed Systems; Steady vs Unsteady State; Multi-component Systems; Chemical Reactions; Problem Solving Strategy	Ch. 3.1–3.2, 2.5	Distinguish system types; formulate material balance problems
M3: Material Balance without Reaction	L7–L9: Material Balance Problems without Reaction	Ch. 4	Solve material balances for single/multi-unit systems
M4: Material Balances Involving Reactions	L10–L12: Stoichiometry & Reaction Terminology; Material Balances with Reaction (Species, Elemental, Combustion)	Ch. 5.1–5.5	Apply stoichiometric relationships in balances
M5: Material Balances For Multi-Unit Systems	L13–L15: Sequential Multi-Unit Systems; Recycle, Bypass & Purge	Ch. 6.1–6.4	Perform balances on interconnected units
M6: Material Balances for Industries	L16–L18: Industrial Applications of Material Balances	Ch. 6.5	Relate material balances to real industrial cases
M7: Ideal gas vs real gas systems	L19–L21: Ideal Gas Law; Ideal Gas Mixtures; Real Gas Compressibility; Real Gas Equations of State	Ch. 7.1, 7.2, 7.3, 7.4	Apply equations of state for gases
M8: Multiphase Equilibrium	L22–L24: Phase Diagram, Vapor Pressure, Phase Rule; Vaporization, Saturation, Condensation	Ch. 8.1–8.3, 8.4	Solve problems with real gases & phase equilibrium
M9: Humidity Charts and Their Use	L25–L27: Humidity Concepts; Humidity Chart	Ch. 11	Calculate humidity & phase-change balances
M10: Energy Balance Preliminaries	L28–L30: Energy Balance Terminology; Types of Energy; Enthalpy Balances (Closed & Open Systems)	Ch. 9	Formulate energy balances for steady/unsteady systems
M11: Energy Balances: How To Account For Chemical Reaction	L31–L33: Heats of Formation, Reaction, and Combustion	Ch. 10.1–10.4	Account for energy effects of reactions
M12: Analysis of the Degrees of Freedom in Steady-State Processes	L34–L36: Degrees of Freedom; Simultaneous Material & Energy Balances	Ch. 12	Analyze & solve integrated material-energy balance problems
M13: Mixing and Dissolution	L37–L39: Heats of Solution and Mixing; Gases & Liquids in Equilibrium with Solids	Ch. 13 & 15	Perform calculations for mixing and dissolution
M14: Unsteady-State Processes	L40–L42: Unsteady-State Material and Energy Balances	Ch. 17	Applying material and energy balances to transient processes



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

7. Evaluation Scheme:

EC No.	Evaluation Component (EC)	Duration (Minutes)	Marks (Weightage)	Date/Time	Remarks
1	Mid-Semester Test	90	90 (30%)	5/10 AN1	CB/OB
2	Comprehensive Examination	180	120 (40%)	3/12 FN	CB
3	Class Participation/Discussion (Linked to attendance)	[In class]	15 (5%)	[Every class]	-
4	Assignments (I & II)	Two Weeks	15 (5%)	-	Take home
5	Tutorial Tests (Best 4 of 6, to be announced beforehand)	-	60 (20%)	-	CB/OB

CB: Close Book OB: Open Book

7. Chamber Consultation Hour: Thursday, 4 to 5 PM. Prior appointment is required for any other time.

8. Notices: All notices concerning this course will be displayed on the **Nalanda** portal.

9. (a) Make-up Policy: Make-up will be granted only when one has genuine reasons (illness requiring hospitalization or similar cases with medical certificate) for not appearing in the regular test. No make-up will be granted for the tutorial tests/surprise class tests/assignments.

(b) Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

10. NC criterion: To clear this course a student has to score at least 20% of total marks i.e., 60 out of 300 marks. Else NC will be awarded.

(Hare Krishna Mohanta)
Instructor-in-Charge
CHE F211